Research on UNet Remote Sensing Road Image Segmentation Method with Attention Mechanism Integration

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***Abstract*—Image semantic segmentation, a key focus in computer vision, holds significant value in remote sensing image analysis, intelligent transportation, and autonomous driving. However, standard UNet models are susceptible to noise in cluttered backgrounds, resulting in imprecise object localization and indistinct boundaries. To overcome this limitation, this paper incorporates an attention mechanism into UNet for segmenting remote sensing road images. Preserving UNet's encoder-decoder architecture and skip connections, we introduce an attention gate that selectively amplifies critical features in skip pathways, enhancing target-relevant information while suppressing extraneous background interference. Experiments on a remote sensing road dataset indicate that this approach surpasses original UNet in mIoU, mPA, accuracy, and Dice score. These findings confirm that the attention mechanism elevates UNet's segmentation accuracy and robustness, offering fresh strategies for precise road and object extraction in remote sensing imagery.**

Keywords-UNet; attention mechanism; remote sensing images

I. INTRODUCTION

Image semantic segmentation is an important research direction in the field of computer vision. Its core task is to accurately classify each pixel in an image, thereby achieving a fine-grained understanding of the scene. Compared with traditional object detection or image classification, semantic segmentation can not only identify what objects are in an image, but also clearly determine the specific location and boundaries of the objects in the image. In recent years, with the development of deep learning, particularly convolutional neural networks, semantic segmentation technology has made significant progress, providing strong support for achieving more intelligent visual perception and human-computer interaction.

Against the backdrop of rapid developments in semantic segmentation technology, research into the semantic segmentation of road images holds significant practical significance and value in applications. With the continuous advancement of smart transportation, autonomous driving, and smart city construction, road environment perception has gradually become a core issue in related fields. Semantic segmentation technology can perform pixel-level fine segmentation of various elements in road scenes, thereby providing a solid technical foundation for a comprehensive understanding of road structures and intelligent applications. Compared to traditional detection and recognition methods, semantic segmentation of road images can precisely depict their spatial distribution and boundary relationships, thereby enabling high-precision modeling and analysis of road scenes[1]. Especially in autonomous driving systems, real-time perception and understanding of road scenes directly impact driving safety and stability. The introduction of semantic segmentation significantly enhances the accuracy and robustness of environmental perception, providing a reliable basis for path planning, obstacle avoidance decisions, and adherence to traffic rules in autonomous vehicles. In the field of intelligent traffic management, road semantic segmentation can assist in the intelligent monitoring of traffic flow, precise analysis of road congestion, and rapid response to emergencies, thereby improving traffic operation efficiency and urban governance levels[2]. Additionally, in application scenarios such as road maintenance and map reconstruction, semantic segmentation technology can automatically identify and update road damage and traffic facility layouts, reducing labor costs and improving work efficiency.

Traditional image segmentation methods include region growing methods, graph cutting algorithms, threshold segmentation methods, and watershed segmentation methods. The primary principle of these algorithms is to divide pixels and then classify them based on similarity. With the successful application of deep learning technology in image classification, many researchers have also focused on using deep learning for image segmentation[3]. Zhongyu Sun et al.[4] addressed the limitations of existing semantic segmentation methods in the extraction of buildings and roads from high-resolution remote sensing images, such as limited receptive fields and loss of detailed information. They proposed a hybrid multi-resolution and Transformer-based semantic extraction network. This method ensures the full retention of high-resolution and multi-scale feature information during downsampling through multi-resolution branch feature fusion, while introducing a Transformer structure to achieve pixel-level global receptive fields, thereby enhancing scene understanding capabilities. Ying Wang et al.[5] addressed the issue of difficulty in extracting small-sized roads in high-resolution remote sensing images due to their elongated shapes and obstruction by vegetation and buildings. They proposed an improved dual decoder UNet (DDUNet) model. This method adds a small decoder to the UNet to obtain more detailed features and introduces a dilated convolution attention module (DCAM) between the encoder and decoder to expand the receptive field and extract multi-scale features through cascaded dilated convolutions and global average pooling. Additionally, a convolutional block attention

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module (CBAM) is embedded in the parallel branches to enhance the focus on important features. Qiming Yang et al.[6] proposed introducing SimAM and CBAM attention mechanisms into the UNet to enhance the model's feature extraction and representation capabilities. Experimental results show that adding SimAM, CBAM, or a fusion of both modules significantly improves segmentation performance. Xingjian Gu et al. [7] addressed issues such as multi-scale targets, large intra-class variations, and imbalanced class distributions in the semantic segmentation of remote sensing images by proposing an adaptive enhanced Swin Transformer and UNet fusion model that combines the advantages of CNN and Transformer. This model adopts a U-shaped encoding-decoding structure based on Transformer and extracts local and global semantic features through skip connections. It introduces an enhanced multi-head self-attention mechanism and a deformable adaptive patch merging layer in the encoder to enhance feature modeling capabilities. In the decoder part, a symmetric cascading structure is designed to achieve high-resolution feature map reconstruction. Yu Sun et al. [8] proposed an improved U-Net based on multi-attention mechanisms (MA-UNet) to address the challenges of insufficient accuracy for multi-class targets and multi-scale processing difficulties in the semantic segmentation of remote sensing images. A residual structure and a simple attention module were introduced in the encoder to enhance the extraction of fine-grained features. A multi-head self-attention mechanism was applied at the lowest feature level to reconstruct semantic representations, thereby improving segmentation performance for pixels of different categories. Sijun Dong et al. [9] proposed BMDANet—a network integrating cross-layer information exchange with Block Multi-Dimensional Attention (BMDA) modules—to address the segmentation of slender road objects in high-resolution remote sensing images (RSI). By constructing a global attention module using multi-dimensional information, this approach optimizes road feature extraction, thereby improving road segmentation performance. To address this, this study introduces an attention mechanism into the UNet structure by embedding attention gate modules in the skip connections to achieve selective enhancement of feature maps. The attention gate can weight and fuse features from the encoder and decoder to generate dynamic attention masks, thereby highlighting regions of interest and suppressing irrelevant information. This design not only effectively enhances the network's sensitivity to target boundaries but also improves the model's robustness and focus in complex backgrounds, providing a new approach for precise segmentation of target regions in remote sensing imagery.

II. Related Theory

A. UNet Network

UNet is a typical encoder–decoder semantic segmentation network, first proposed and applied to medical image segmentation by Ronneberger et al.[10]. Its overall structure is symmetrical and U-shaped, consisting mainly of an encoder, a decoder, and skip connections. The encoder gradually reduces the spatial resolution of feature maps through multiple layers of convolution and pooling operations, thereby extracting high-level abstract semantic information. The decoder gradually restores the spatial resolution through deconvolution or upsampling operations, enabling pixel-level fine-grained predictions.

The core design of the UNet is the skip connection between the encoder and decoder. This mechanism directly passes high-resolution features from lower layers of the encoder to corresponding layers of the decoder, where they are fused with upsampled features. This effectively preserves local details and edge information, compensating for spatial information loss caused by multiple downsampling operations. With this structure, UNet achieves global semantic understanding while maintaining clear object boundaries and complete details, significantly improving segmentation accuracy. Its network structure is shown in Figure 1.

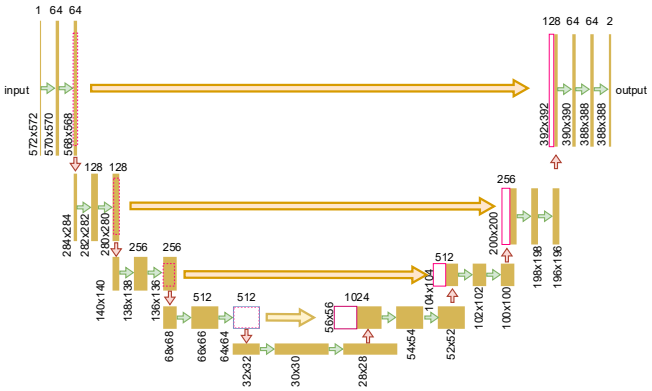


Figure 1. UNet network structure

B. Attention Mechanism

The attention mechanism originates from the human visual cognitive process, with its core idea being to simulate how humans focus on key information areas while ignoring irrelevant or secondary information when perceiving information. In deep learning, the attention mechanism assigns different weights to input features, enabling the model to automatically learn "which parts should be given more attention." The attention mechanism typically generates weight distributions by calculating the correlations between queries, keys, and values, and applies these weights to feature maps to enhance important features and suppress irrelevant ones. This mechanism enhances the model's ability to focus on target regions and improve modeling of long-range dependencies, thereby addressing the limitation of traditional convolutional neural networks, which only have local receptive fields.

In computer vision, the attention mechanism has been applied to tasks such as image classification, object detection, and semantic segmentation, with common forms including Channel Attention, Spatial Attention, and Self-Attention. In image semantic segmentation, the attention mechanism guides the model to focus on prominent features in the target region while suppressing interference from complex backgrounds, thereby effectively improving segmentation accuracy and robustness.

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III. Improvements to UNet based on Attention Mechanisms

This study introduces an attention mechanism at the skip connection to filter features passed from the encoder. Unlike existing channel attention (e.g., SE, CBAM) or self-attention (e.g., Transformer) approaches, the proposed Attention Gate (AG) does not rely solely on channel weights or global dependency modeling. Instead, it performs spatially selective filtering at the skip connection of the U-Net architecture. AG simultaneously utilizes high-level features as guidance signals and low-level features. Through a gating mechanism, it dynamically generates weight masks, enabling cross-level feature interaction and selection. This mechanism allows the model to automatically suppress redundant background information during the concatenation stage while highlighting task-relevant regions, thereby balancing global semantic understanding with the preservation of local details.

The feature maps from the decoder and encoder are first reduced in dimension via convolution and BatchNorm, respectively. They are then added together, passed through a ReLU activation function, and finally combined with a 1×1 convolution and Sigmoid function to generate an attention mask. This mask is multiplied pointwise with the encoder features to emphasize regions relevant to the target and suppress irrelevant regions[11].

In traditional UNet, encoder features are directly concatenated with the decoder via skip connections. In Attention UNet, this process is illustrated as Figure 2.

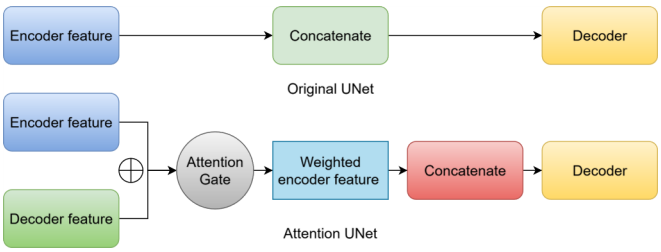


Figure 2. Attention UNet encoder and decoder concatenation method

The module is placed between the "jump connection" and "decoder concatenation" to perform "selective filtering" on the feature maps passed over, avoiding interference from irrelevant features.

The mathematical calculation process of the attention gate is as follows:

- Input decoder feature g and encoder feature x. $g \in \mathbb{R}^{B\times F_g\times H_g\times W_g}$, $x \in \mathbb{R}^{B\times F_l\times H_x\times W_x}$.
- Map the features to a low-dimensional space, use 1×1 convolution and BN to reduce the number of channels to F_{int}, so that g' and x' have the same dimensions, which is convenient for addition.

$$x' = W_x(x) \in \mathbb{R}^{B\times F_{int}\times H\times W} \tag{1}$$

$$g' = W_g(g) \in \mathbb{R}^{B\times F_{int}\times H\times W} \tag{2}$$

- The two are combined for feature fusion, then processed by ReLU.

$$\psi' = \mathrm{ReLU}(g' + x') \tag{3}$$

- Generate an attention mask, then apply a 1×1 convolution, BN, and Sigmoid compression to a single channel.

$$\psi = \sigma\left(W_\psi(\psi')\right) \in \mathbb{R}^{B\times 1\times H\times W} \tag{4}$$

Where σ is the Sigmoid activation function, ensuring that the weights are in the range [0,1].

- Finally, the encoder features are adjusted using the attention mask ψ, which is multiplied pixel by pixel to give the target area a higher weight and the background area a weight close to 0.

$$x_{\mathrm{att}} = x \cdot \psi \tag{5}$$

Summary of overall calculation formula.

$$x_{\mathrm{att}} = x \cdot \sigma\left(W_\psi\left(\mathrm{ReLU}\left(W_g(g) + W_x(x)\right)\right)\right) \tag{6}$$

The structure of the UNet network incorporating the attention mechanism is shown in Figure 3. The advantage of this structure is that it significantly improves segmentation accuracy and robustness through the attention mechanism, especially when the target area is small or the boundaries are complex. It not only enhances the response to areas of interest but also effectively reduces interference from background areas, making it suitable for various image segmentation tasks with high precision requirements.

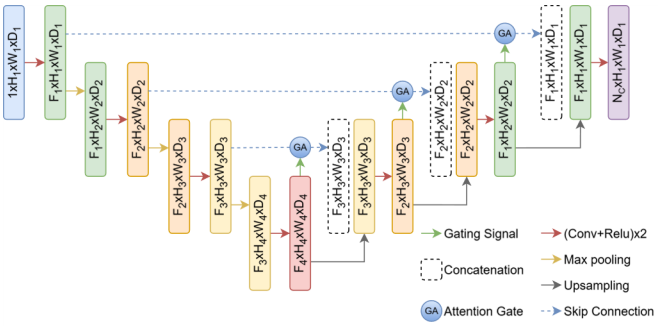


Figure 3. UNet network structure incorporating attention mechanisms

IV. Experimental Design and Results Analysis

A. Evaluation Criteria

Evaluation metrics in semantic segmentation are primarily used to measure the performance of a model during the prediction process. Commonly used evaluation metrics include precision, recall, mean intersection over union (mIoU), mean pixel accuracy (mPA), and Dice coefficient[12].

- Precision and Recall primarily measure the model's performance in predicting positive classes from different perspectives. One focuses on reducing false

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positives, while the other focuses on reducing false negatives.

$$P = \frac{TP}{TP+FP} \tag{7}$$

$$R = \frac{TP}{TP+FN} \tag{8}$$

- mIoU calculates the degree of overlap between the predicted and actual labels, with higher values being better.

$$IoU_i = \frac{TP_i}{TP_i+FP_i+FN_i} \tag{9}$$

$$mIoU = \frac{1}{C}\sum_{i=1}^{C} IoU_i \tag{10}$$

- mPA focuses on pixel-level accuracy for each category and is suitable for multi-category segmentation tasks.

$$PA_i = \frac{TP_i}{TP_i+FN_i} \tag{11}$$

$$mPA = \frac{1}{C}\sum_{i=1}^{C} PA_i \tag{12}$$

- The Dice coefficient is a comprehensive indicator for measuring the similarity between two areas.

$$Dice = \frac{2*TP}{2*TP+FP+FN} \tag{13}$$

B. Experimental Data

The experimental data used in this study were obtained from the Internet. The original images and corresponding mask images were divided into a training set and a test set in a ratio of 7:3. Figure 4 shows examples of experimental remote sensing road images, with the original image on the left and the corresponding mask image on the right.



Figure 4. Remote sensing road image examples

C. Experimental Results and Analysis

This experiment was conducted on the original model and the model incorporating the attention module. The training was conducted for 50 rounds with a batch size of 4 and a learning rate of 1e-4. Figure 5 shows the loss changes of the attention UNet model. After training, the loss of the training set basically converged.

The model obtained from training was tested on the test set, and the IoU, Recall, and Precision test results for the background and road parts are shown in Table 1.

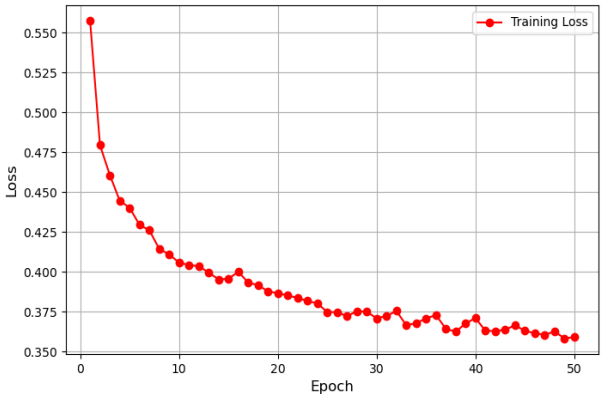


Figure 5. Convergence of training set loss

TABLE I. COMPARISON OF IOU, RECALL, AND PRECISION RESULTS

Model	Background			Road		
	IoU(%)	R(%)	P(%)	IoU(%)	R(%)	P(%)
Original UNet	94.82	98.59	96.12	76.94	81.88	92.72
UNet+Attention	95.35	98.37	96.88	79.64	85.55	92.02

The comparison results of mIoU, mPA, accuracy, and Dice score obtained from different models tested on the test set are shown in Table 2.

TABLE II. COMPARISON OF MIOU, MPA, ACCURACY AND DICE SCORE

Model	mIoU(%)	mPA(%)	Accuracy(%)	Dice score (%)
Original UNet	85.88	90.24	95.58	72.596
UNet+Attention	87.49	91.96	96.06	74.38

As can be seen from the test results, the overall performance of UNet in the road image segmentation task has been significantly improved after incorporating the attention mechanism. Specifically, in terms of category-level metrics, the IoU, Recall, and Precision of the background region have seen slight improvements compared to the original UNet, while the improvements in the road region are more pronounced, with IoU increasing from 76.94% to 79.64%, and Recall from 81.88% to 85.55%, indicating that the attention mechanism effectively enhances the model's focus on the target region (roads) and reduces the false negative rate. In terms of overall metrics, the UNet with attention outperforms the original UNet in mIoU, mPA, Accuracy, and Dice score, with improvements of 1.61%, 1.72%, 0.48%, and 1.78%, respectively, further validating the role of the attention mechanism in optimizing segmentation boundaries and suppressing background interference. Therefore, it can be concluded that the attention mechanism helps the model maintain high background segmentation accuracy while better extracting and utilizing road region features, thereby improving overall segmentation performance.

V. CONCLUSIONS

This paper addresses the limitations of traditional UNet in remote sensing image segmentation, such as insufficient focus on target regions, blurred boundaries, and susceptibility to

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background noise interference. It proposes a remote sensing road image segmentation method that integrates an attention mechanism. By introducing an attention gate module in the skip connections, the method effectively achieves selective feature enhancement, enabling the model to concentrate on target-related areas while reducing interference from redundant background features. Experimental results validate the attention mechanism's advantages in optimizing segmentation performance and enhancing model robustness, providing reliable technical support for applications such as intelligent transportation, autonomous driving, and urban planning.

However, this study still has certain limitations, such as sensitivity to noise and insufficient adaptability to data from different sensors and resolutions. Future research may explore integrating structures like Transformers, adaptive convolutions, or graph neural networks to enhance modeling capabilities for global dependencies and multi-scale features. Alternatively, lightweight network designs could be investigated to improve the model's applicability in resource-constrained environments. Additionally, incorporating multimodal fusion techniques could boost the model's generalization performance on heterogeneous remote sensing data.

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